UNIVERSITY OF CAMBRIDGE | Department of Engineering

Fission Matrix Acceleration in Monte Carlo Simulations and Its Impact on Neutron Clustering

Phase 1 Report by

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In Partial Fulfillment of the Requirements for the Degree of
Master of Philosophy in Nuclear energy

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Submitted on

17 March 2026

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Contents

1 Introduction

Monte Carlo neutron transport simulations are a widely used method for computing the behavior of nuclear systems. Their benefit compared to other transport methods results from the low number of approximations that are made about the underlying physics. This lack of simplifications means that Monte Carlo derived results are highly accurate. However, the advantage in accuracy comes at the expense of computation time. The stochastic nature of these simulations means that a large number of neutrons need to be simulated for many cycles before the quantities converge. For that reason, the reduction of variance in Monte Carlo cycles is an important area of research in computational neutronics.

In recent years, it has become known that Monte Carlo simulations exhibit inter-generational spatial correlations due to geometric asymmetries in the birth and death of neutrons. Such correlations can manifest themselves as persistent non-physical variations in local neutron densities that increase the variance of the fission source. This phenomenon is known as neutron clustering. Prior attempts to address this issue have led to marginal improvements whose effectiveness is limited to idealized systems. A comprehensive remedy remains elusive.

This project will attempt a novel solution using a method known as fission matrix acceleration. This technique has previously been employed to accelerate the convergence of the fission source, but its effect on neutron clustering has not been properly studied. The following report will start by introducing the Monte Carlo transport method, as well as the problem of clustering. This will be followed by an explanation of fission matrix acceleration and a theoretical hypothesis of its beneficial effect on clustering. To study the validity of this claim, two numerical experiments will be proposed. The report will conclude with a planning for the next phase of this project. This will include some optional extensions in case additional time is available.

2 Monte Carlo Neutron Transport

The following section discusses the most widely used Monte Carlo (MC) neutron transport method. This method relies on applying the power iteration algorithm to the k-eigenvalue equation[1,2].

2.1 k-Eigenvalue Equation

Consider a material filled with neutrons. Let ϕ denote the scalar neutron flux field in the material. Each generation of neutrons is governed by several interactions: Neutrons can scatter, fission, leak out, or be captured. These interactions are modeled using their respective linear operators: the *scatter operator* $\mathbf{S}$, the *fission multiplication operator* $\mathbf{F}$, and the *leakage operator* $\mathbf{L}$. Capture is incorporated in the *collision operator* $\mathbf{T}$, which represents scattering, fission, and capture.

When a system is critical, the neutrons produced in each generation should balance out the neutrons that are absorbed:

$$[\mathbf{L} + \mathbf{T}]\phi = [\mathbf{S} + \mathbf{F}]\phi. \tag{1}$$

However, this equality does not hold for subcritical or supercritical systems. This can be accounted for by including a balancing factor k. This *effective multiplication factor* can be interpreted as the growth in the number of neutrons in a generation. Adding k to (1) gives:

$$[\mathbf{L} + \mathbf{T} - \mathbf{S}]\phi = \frac{q}{k}, \tag{2}$$

where q is the *fission source* defined by $q = \mathbf{F}\phi$. (2) is the *k-eigenvalue equation.* The purpose of the MC algorithm is to find an eigenpair k and q such that (2) holds.

It is useful to have elementary analytical expressions for the linear operators. To this end, consider a simplified system with one-dimensional slab geometry and mono-energetic neutrons. This allows most operators to be replaced with macroscopic cross-sections: $\mathbf{T} = \Sigma_\mathrm{t}(x)$, $\mathbf{S} = \Sigma_\mathrm{s}(x)$, and $\mathbf{F} = \bar{\nu}(x)\,\Sigma_\mathrm{f}(x)$. The symbols Σ_t, Σ_s, and Σ_f respectively denote the macroscopic total, scattering and fission cross-sections; $\bar{\nu}$ is the neutron yield. As for $\mathbf{L}$, it is possible to make the diffusion approximation: $\mathbf{L} = -\frac{\mathrm{d}}{\mathrm{d}x}\left[D(x)\frac{\mathrm{d}}{\mathrm{d}x}\right]$, where $D(x)$ is the diffusion coefficient. This approximation is known to be accurate for large homogeneous regions that are not highly absorbent or anisotropic[2]. Rewriting (2) gives:

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left[D(x)\frac{\mathrm{d}\phi(x)}{\mathrm{d}x}\right] + \Sigma_\mathrm{a}(x)\phi(x) = \frac{\bar{\nu}(x)\Sigma_\mathrm{f}(x)\phi(x)}{k}, \tag{3}$$

where Σ_a is the macroscopic absorption cross-section.

2.2 Monte Carlo Power Iteration

As stated before, the MC algorithm uses power iteration to find an eigenpair of (2). Specifically, the power iteration method converges to the largest absolute eigenvalue. In neutron transport, this is referred to as the *fundamental mode* with eigenvalue k_0. Smaller eigenvalues are successively denoted $k_1,\ k_2,\ k_3$, etc. The fundamental-mode fission source is denoted q_0. The *dominance ratio* is defined by $R = k_1/k_0$. It is known that lower values of R tend to converge faster[3].

The operators in (2) are modeled by simulating the physical interactions they represent, such that each neutron generation represents a power iteration. For MC simulations, these successive generations are referred to as *cycles*. Algebraically:

$$[\mathbf{L} + \mathbf{T} - \mathbf{S}]\phi^{(n+1)} = \frac{q^{(n)}}{k^{(n)}}, \tag{4}$$

where the superscript corresponds to the generation. In essence, each generation n simulates the operation $\mathbf{F}\,[\mathbf{L} + \mathbf{T} - \mathbf{S}]^{-1} q^{(n)}/k^{(n)}$ in order to find $q^{(n+1)}$. Next, $k^{(n+1)}$ is determined by calculating the ratio by which the number of neutrons has increased:

$$k^{(n+1)} = \frac{\iiint q^{(n+1)} \mathrm{d}V}{\frac{1}{k^{(n)}} \iiint q^{(n)} \mathrm{d}V}. \tag{5}$$

This process is repeated until the eigenpair, or some other derived quantity, has converged.

2.3 Algorithm

A detailed flowchart of the MC algorithm is shown in Figure 1. In short, the algorithm starts with an initial guess for $q^{(0)}$ and $k^{(0)}$. Then, each neutron is moved to a random location where it undergoes an interaction. If the interaction is a scatter, the neutron is moved again to another interaction. If instead the interaction is a capture or a fission, the neutron is *killed*. Additionally in the case of fission, a random number of new *daughter neutrons* are created in a *fission bank*. This process is repeated until all neutrons have been killed. This completes a cycle.

Before starting the next cycle, the neutrons in the fission bank must be *normalized* such that the number of neutrons stays constant in every cycle. This is accomplished by duplicating a random selection of neutrons (*splitting*) or removing them (*Russian roulette*). After normalization, the fission bank becomes the fission source for the next cycle $q^{(n+1)}$. The aforementioned operation is repeated for multiple cycles until the eigenpair approaches the fundamental mode. These are known as the *inactive cycles*, because no data is collected during this period. The *active cycles* are performed after the inactive cycles have finished. During the active cycles, the quantity of interest is tallied. The result is calculated by averaging the tallies of a large number of active cycles.

3 Neutron Clustering

The phenomenon of *neutron clustering* was first coined by Dumonteil et al.[4] to describe spatial and generational correlations between neutrons in MC simulations. When the number of simulated neutrons is low, there is a tendency for some regions to have a higher local neutron density in a way that is inconsistent with q_0. In other words, such high-density regions are statistically improbable for larger numbers of neutrons. These regions are known as *clusters*. Neutron clustering is a form of *neutral clustering*, meaning that the correlations arise in the absence of any interactions between neutrons. Instead, the correlations are a result of the asymmetry between two processes: the birth and death of neutrons. Neutrons are born through fission or splitting, which must happen near other neutrons. In contrast, neutrons die due to capture or Russian roulette, which happens more uniformly through space. The combination of these two effects has the tendency to push successive generations of neutrons closer together.

Sutton and Mittal[5] found that after a certain amount of cycles, all neutrons will have one common ancestor. This moment is known as *fixation*, and the generation when this happens is the *fixation generation*. In the absence of movement, all neutrons would be located at the same point after the fixation generation. However, this tendency is counteracted by the spatial diffusion of neutrons. Clustering only occurs when diffusion is too weak to negate the correlations from the fixation process. For finite systems, the generation by which a neutron has explored the whole space is known as the *mixing generation*. Clustering tends to happen when the mixing generation approaches or exceeds the fixation generation. It has been shown that this is inversely proportional to the neutron density. Therefore, lower neutron densities lead to more clustering.

Neutron clustering has several adverse effects. Nowak et al.[6] showed that the Shannon entropy of $q^{(n)}$ still converges in the presence of neutron clustering, albeit to a lower value. This may give the false impression that $q^{(n)}$ has spatially converged q_0, even when neutron clusters persist indefinitely. More importantly, the presence

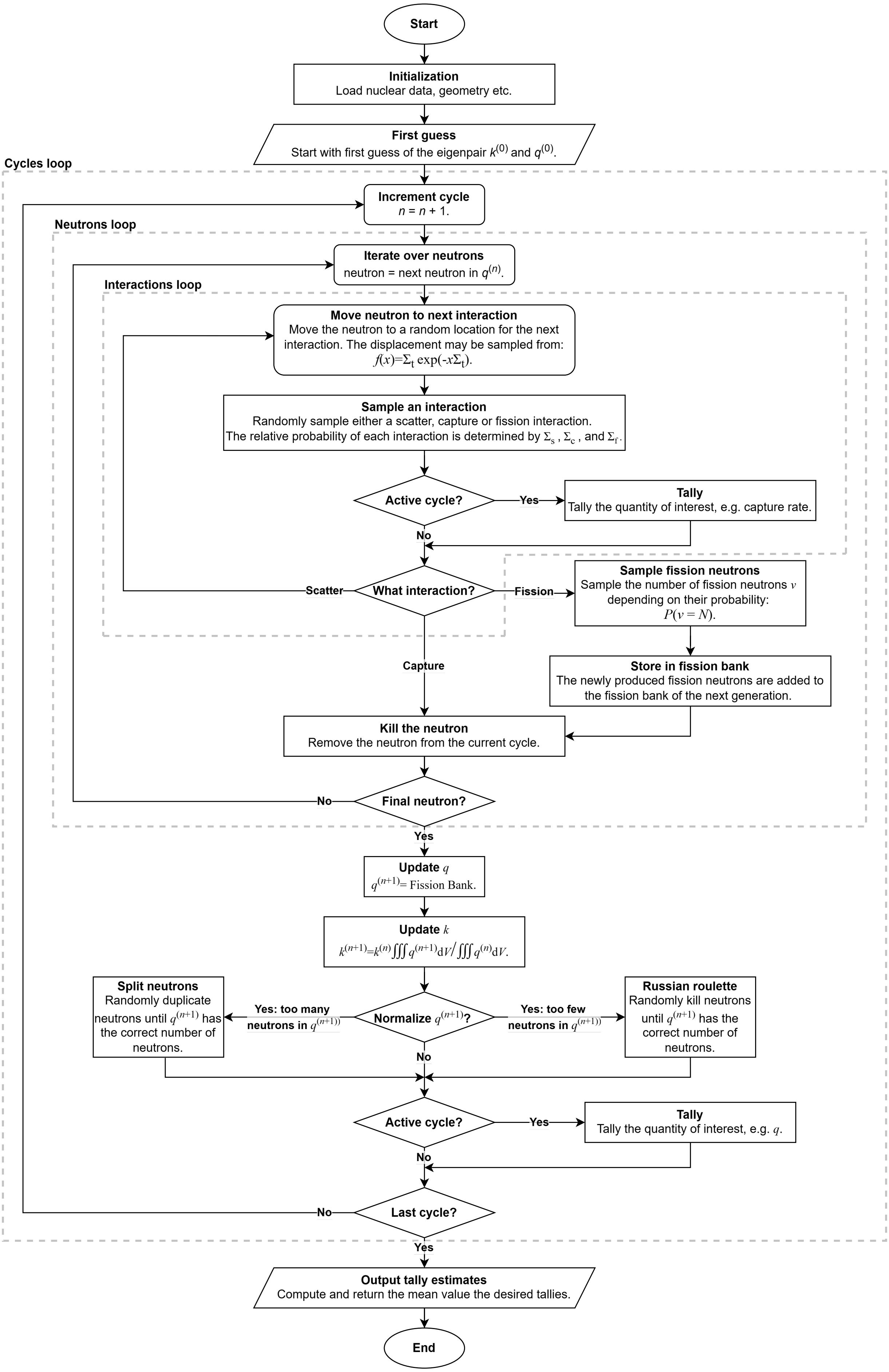

Figure 1: Flowchart for the k-eigenvalue Monte Carlo power iteration algorithm.

of these clusters slows the convergence of the tally-averaged q, and all quantities derived from it. Nonetheless, the tally-averaged q does correctly converge to q_0.

Mickus and Dufek[7] have demonstrated that, over many generations, neutron clustering has no adversarial effect on the convergence or results of Monte Carlo tally counts. That is to say: neutron clustering only effects local quantities.

It should be noted that clustering is only problematic when the number of simulated neutrons is much lower than the actual number it is trying to model. This does not apply to cases where the real number of neutrons low aswel. Physical clustering has been observed in such cases[8].

3.1 Mitigation

Various methods to reduce the effect of neutron clustering have been proposed. The most well-researched of these involve changing the *branching* algorithm to a *branchless* algorithm. A normal MC cycle as described in subsection 2.3 is a branching algorithm, because single neutrons produce multiple daughter neutrons during fission. Neutron clustering happens because these daughter neutrons are spatially correlated to their parent. A branchless algorithm prevents this by only creating one daughter neutron per fission. Instead, the creation of new neutrons is modeled by increasing their statistical weight. Belanger et al.[9] proposed increasing the weight at every interaction by a factor of $(\Sigma_\mathrm{s} + \bar{\nu}\Sigma_\mathrm{f})/\Sigma_\mathrm{t}$. On the other hand, Sutton[10] suggested increasing the weight at every fission by a factor of $\bar{\nu}\Sigma_\mathrm{f}/\Sigma_\mathrm{a}$. Branchless methods have been shown to successfully reduce the effect of clustering[11].

One problem that is not addressed by branchless methods is the normalization step. The neutron population needs to be kept constant, but the usual method of randomly splitting or removing neutrons still introduces correlations. Multiple solutions have been proposed to address this. Notable ones include the sampling without replacement[12], and the adaptive multilevel splitting[13] algorithms. Combining these improved population control methods with the branchless algorithms has been shown to further reduce neutron clustering[11,14]. Nevertheless, some amount of neutron clustering still persists, especially in more realistic systems. For a reactor core, Bonnet and Belanger[15] have found that branchless methods are less beneficial and sometimes even counterproductive.

Lastly, Cosgrove et al.[16] have suggested an entirely different method that uses the fission matrix to estimate q_0. This estimated fission source is subsequently used to redistribute neutrons in every $q^{(n)}$. This project will be studying a modified version of this method. A detailed explanation for this will be given in section 4.

4 Fission Matrix Acceleration

A *fission matrix* (FM), denoted $\mathbf{H}$, can be used to accelerate the convergence of $q^{(n)}$[17–19]. First, the space is divided into regions, or *bins*. The value of H_{ij} represents the number of neutrons produced in bin i, by the neutrons in bin j, normalized to the number of neutrons in bin j. The geometry of these bins can be arbitrary, although the resolution is typically low. The value of H_{ij} can be thought of as the local effective multiplication factor of bin i from neutrons in bin j. It logically follows that q_0 must satisfy the equation:

$$\mathbf{H}q_0 = k_0 q_0, \tag{6}$$

where q_0 must be downsampled to match the FM bins. From (6), it can be seen that q_0 is equal to the fundamental eigenvector of $\mathbf{H}$, denoted q_FM.

The fact that H_{ij} is normalized to the number of neutrons in bin j, means that it can be estimated without an accurate $q^{(n)}$. Therefore, it is possible to evaluate $\mathbf{H}$ during the each cycle. The neutrons population in the different bins of $q^{(n)}$ can then be rescaled to match q_FM through splitting and Russian roulette. This whole procedure is known as *fission matrix acceleration*. Since q_FM is derived from $\mathbf{H}$, it contains information about the relation between any pair of regions, irrespective of the distance between them. It therefore accelerates the $q^{(n)}$ convergence for larger spaces, because information can travel non-locally. Note that the low resolution of $\mathbf{H}$ means that it is not computationally intensive to determine.

Terlizzi and Kotlyar[20] have found that there exists an optimal resolution for the FM bins: Higher or lower resolutions result in a slower convergence. This optimal value is different depending on if the convergence of $q^{(n)}$ or $k^{(n)}$ is being optimized. Regardless, the resolution ends up being quite sparse for both cases. In fact, it is noted that high resolutions can actually worsen neutron clustering compared to not using the FM. This is because the smaller bin sizes means that there are not enough neutrons in each bin to accurately determine H_{ij}.

4.1 Hypothesis: Impact on Clustering

FM acceleration is typically used to accelerate the convergence of $q^{(n)}$. However, it may be possible to use FM acceleration to reduce the effect of neutron clustering. As explained above, q_FM is an estimation of q_0 that is not based on localized behavior. Neutron clusters are local deviations from q_0. Therefore, it is hypothesized that the same FM acceleration procedure can reduce neutron clusters in $q^{(n)}$. Cosgrove et al.[16] have previously found some limited success using a similar method. The difference is that their method replaces a fixed fraction of neutrons in $q^{(n)}$ by neutrons sampled from q_{FM}. In contrast, FM acceleration rescales the number of neutrons in each bin depending on the difference between $q^{(n)}$ and q_{FM}.

The idea is reliant on q_{FM} being an accurate enough estimate of q_0 such that clusters are indeed reduced. Specifically, if the resolution is too low compared to the cluster size, there might be no benefit at all. Additionally, as discussed before, a resolution that is too high can worsen neutron clustering. Nevertheless, no comprehensive studies on the relation between the FM resolution and clustering exist. The goal of this project is to analyze if and how FM acceleration effects neutron clustering at different resolutions.

5 Shannon Entropy

In order to study neutron clustering, it is first necessary to quantify it. A variety of estimators for the level of clustering can be found in the literature[4,5,11,21]. One method proposed by Nowak et al.[6], is to look at the *Shannon entropy*. A comparison between different methods by Bonnet et al.[11] found that Shannon entropy is a computationally inexpensive estimator when detailed spatial information is unnecessary. Since this study is limited to simple systems, Shannon entropy will be used for all clustering measurements.

5.1 Definition

Consider a one-dimensional space filled with neutrons. This space is divided into B smaller *cells*. Shannon entropy $\mathcal{S}$ is defined as:

$$\mathcal{S} = -\sum_{i\in B} p_i \log_2(p_i), \tag{7}$$

where p_i denotes the fraction of neutrons that are inside of cell i. For a homogeneous system with a finite number of neutrons N, the Shannon entropy is given by:

$$\mathcal{S}_N = \log_2(N) - \frac{B^{1-N}}{N} \sum_{K=0}^{N} \binom{N}{K} (B-1)^{N-K}\, K \log_2(K). \tag{8}$$

It is known that $\mathcal{S}$ converges after some number of generations[6]. However, as stated before, this does not necessarily imply that $q^{(n)}$ has converged to q_0. The amount by which the converged value of $\mathcal{S}$ is lower than the analytically determined $\mathcal{S}_N$ indicates the degree of persistent neutron clustering. This is how $\mathcal{S}$ is used as a clustering estimator.

5.2 Interpretation

Entropy itself is a general statistical concept that has many interpretations across different fields. A particularly useful interpretation comes from statistical physics, where entropy is seen as a measure for the number of microstates per macrostate. It is known that a sufficiently large number of neutrons will naturally find the fundamental-mode distribution. This is because there are many microscopic configurations of neutrons that correspond to the fundamental mode. In contrast, a particular neutron cluster demands that neutrons are distributed in such a way that the cluster is present. There are fewer configurations of neutrons that are consistent with this, therefore the entropy is lower. This interpretation naturally explains why neutron clusters only exist for small numbers of neutrons.

Another interpretation comes from information theory. Entropy can be seen as a measure for the amount of information carried by a signal. Since distributions with clusters correspond to fewer microstates, they carry less information about the ensemble of all possible states. This interpretation is an intuitive way to understand why neutron clustering leads to slower convergences.

6 Methodology

The influence of FM acceleration on neutron clustering will be tested using numerical experiments. This will involve performing $k-$eigenvalue MC simulations on two different setups. These setups are analytically solvable, which is useful for analyzing the results. Simulations are done with mono-energetic neutrons and isotropic scattering in one-dimensional slab geometry. All simulations will be performed using SCONE, a neutron transport code developed by the Nuclear Energy group at the University of Cambridge[22].

6.1 F-A-F setup

The first setup consists of 3 regions: an absorbing region in between two fissile regions on both sides. The fissile regions will only have fission and scatter interactions, such that $\Sigma_a = \Sigma_f$. On the other hand, the absorbing region will only have capture and scatter interactions, so $\Sigma_a = \Sigma_c$. The cross-sections are set to $\Sigma_t = 1$, $\Sigma_s = 0.99$, and $\Sigma_a = 0.01$ for all regions. The absorbing region has length a, while the whole setup has length L. The setup is centered around $x = 0$. The boundary conditions are reflective. A diagram of the setup is shown in Figure 2.

Since the absorbing region cannot produce neutrons, it follows that most of the neutrons will be located in the two fissile regions. The symmetry of the setup means that the number of neutrons in these regions should be identical. When a cluster develops, the number of neutrons in the two regions is no longer equal. The absorbing region in the middle makes it very difficult for this cluster to disperse naturally. This is because neutrons need to cross the entire absorbing region to balance the populations of the fissile regions. Thus, the cluster will likely persist into the next generation.

This setup is a favorable case for the fission matrix: The values in $\mathbf{H}$ are not dependent on the number of neutrons in each region, so q_{FM} should be symmetric. By rescaling each region to q_{FM}, it may be possible to mitigate neutron clustering. Since there are only 2 fissile regions, only 2 FM bins are needed: one for each region.

Multiple simulations will be performed both with and without FM acceleration. Different numbers of neutrons will be simulated to study the effect of neutron density. The level of clustering will be determined by comparing $\mathcal{S}$ to the theoretical value. The resulting flux distributions will be compared to the analytical ϕ to gauge the accuracy of the simulation results. It is hypothesized that FM accelerated simulations will exhibit less clustering and converge faster. The analytical expressions for ϕ and $\mathcal{S}_N$ are derived in the next section.

6.1.1 Analytical Solution

The neutron diffusion equation (3) will be used to find the analytical neutron distribution. Since scattering is isotropic, the diffusion coefficient is given by: $D = 1/(3\Sigma_t)$. The absorbing and fissile regions will be considered separately.

Starting with the absorbing region $|x| < a/2$, the fission cross-section can be eliminated. Substituting $\Sigma_f = 0$ and $\Sigma_a = \Sigma_c$ into (3) gives:

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = \frac{\phi}{L_D^2}, \quad L_D = \sqrt{\frac{D}{\Sigma_c}}. \tag{9}$$

The setup is symmetric, so the solution must be even. Thus, the solution is:

$$\phi = A\cosh(x/L_D) \quad \text{for } |x| < a/2, \tag{10}$$

where A is some scaling constant. Note that there is no odd sinh component. As for the fissile region $a/2 < |x| < L/2$, there is only fission and no capture. Changing $\Sigma_c = 0$ and $\Sigma_a = \Sigma_f$ in (3) results in:

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = -\kappa^2\phi \quad , \quad \kappa = \sqrt{\frac{(\bar{\nu}/k - 1)\Sigma_f}{D}}. \tag{11}$$

Note that κ is real because $\bar{\nu}/k - 1 > 0$. The solution is found by imposing reflective reflective boundary conditions around $x = L/2$:

$$\phi = B\cos(\kappa[x - L/2]) \quad \text{for } a/2 < x < L/2. \tag{12}$$

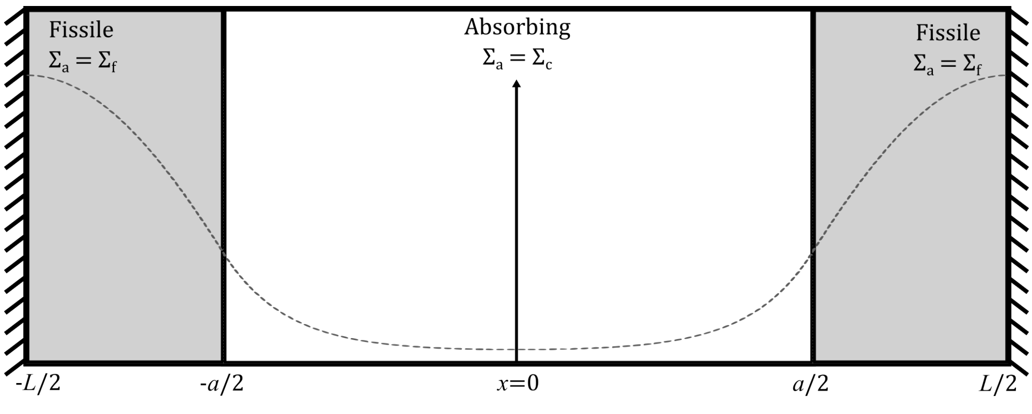

Figure 2: Diagram of the F-A-F setup. The vertical arrow indicates the origin of the x-axis. The left and right sides have reflective boundary conditions. The dashed line is the analytical solution of the neutron flux distribution.

An expression for B can be found by noting that ϕ is continuous at $a/2$.

$$B = \frac{A\cosh\left(a/[2L_D]\right)}{\cos\left(\kappa[a-L]/2\right)}. \tag{13}$$

For the diffusion approximation, the neutron current is the spatial derivative of ϕ multiplied by $D(x)$. The current is continuous and $D(x)$ is constant, so ϕ must be smooth around $a/2$:

$$\left.\frac{\mathrm{d}\phi_{\text{absorb}}}{\mathrm{d}x}\right|_{x=a/2} = \left.\frac{\mathrm{d}\phi_{\text{fissile}}}{\mathrm{d}x}\right|_{x=a/2}. \tag{14}$$

Solving (14) and substituting (13) gives the transcendental equation:

$$\kappa\tan(\kappa[a-L]/2) = -\tanh(a/[2L_D])\,/L_D. \tag{15}$$

(15) can be used to find solutions for κ. The fundamental-mode eigenvalue k_0 corresponds to the solution with the smallest κ. Therefore, the fundamental-mode flux distribution becomes:

$$\phi = \begin{cases} \dfrac{A\cosh\left(a/[2L_D]\right)}{\cos\left(\kappa[a-L]/2\right)}\cos\left(\kappa[x+L/2]\right), & \text{for } -L/2 < x < -a/2 \\ A\cosh\left(x/L_D\right), & \text{for } |x| < a/2 \\ \dfrac{A\cosh\left(a/[2L_D]\right)}{\cos\left(\kappa[a-L]/2\right)}\cos\left(\kappa[x-L]/2\right), & \text{for } a/2 < x < L/2 \end{cases}. \tag{16}$$

Given the dimensions of $a = 60$ and $L = 100$, the following quantities are found: $L_D = 5.774$, $k_0 = 0.8881\bar{\nu}$. The resulting distribution is shown in Figure 2.

The analytical value of $\mathcal{S}_N$ is found using (8). This equation is only valid for homogeneous systems. Nevertheless, this condition can be fulfilled by only using 2 cells corresponding to the positive and negative axes. The symmetry of the system means that these 2 regions are equal. Thus, the analytical expression for $\mathcal{S}_N$ is:

$$\mathcal{S}_N = \log_2(N) - \frac{2^{1-N}}{N}\sum_{K=0}^{N}\binom{N}{K}K\log_2(K). \tag{17}$$

The low spatial resolution of $\mathcal{S}$ is not a problem for this setup. This is because neutron clustering manifests as an asymmetry in the number of neutrons in the two fissile regions. Therefore, one cell corresponding to each region is enough to measure this inequality.

6.2 Homogeneous setup

The second setup is a homogeneous space with periodic boundary conditions. The analytic neutron distribution for this trivial case is constant in space. Any localized density variation is a form of clustering.

This setup is used because it represents an alternative scenario for the FM. There will be clusters of many different sizes throughout the system. How FM acceleration will effect these clusters will likely depend on the resolution, which may neither be too high nor too low. The number of neutrons also plays a role, as this determines the neutron density and by extension the amount of clustering. Unlike the F-A-F setup, simulations will need to vary both the number of neutrons, and the resolution of $\mathbf{H}$. It is hypothesized that there exists some optimal resolution that minimizes the level of clustering.

6.3 Extensions

The research can be extended in case additional time is left over. The first extension would be to study the theoretical causes behind the results of the numerical experiments. Obviously, some surface level theoretical analysis must necessarily be part of the dissertation. The extension would be to derive the experimental observations using theory.

The second broader extension would be to perform additional simulations by modifying the two setups. For example, it would be possible to vary the length of the absorbent region in the F-A-F case. Another idea is to increase the number of fissile regions. It is also possible to change the cross-sections of the materials, which is equivalent to changing the scale of the systems.

7 Planning

A Gantt chart of the planning for phase 2 of this project is shown in Figure 3. It is divided into columns of 'Must', 'Should', and 'Could' depending on the importance of their completion. The planning is designed such that only the 'Must' column is strictly necessary to finish the project without any extensions. Supervisor meetings will happen on Fridays, which are indicated by the major axis.

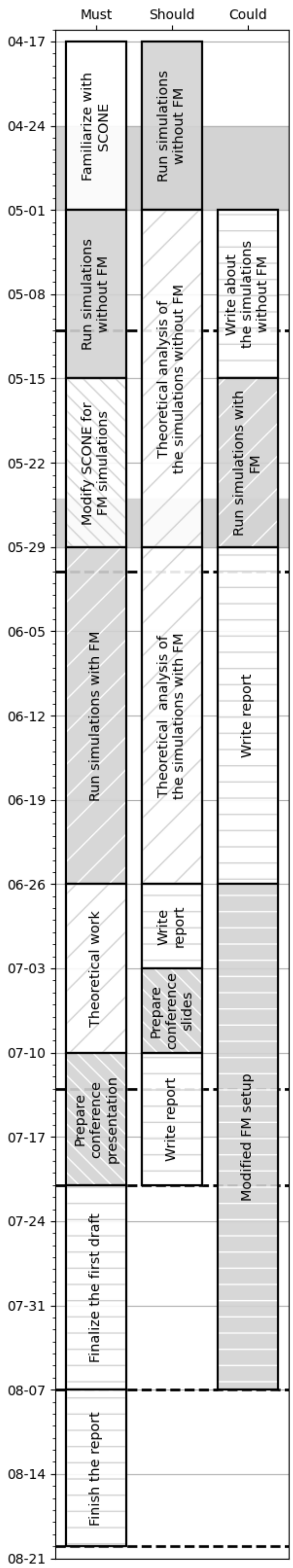

Figure 3: Gantt chart.

- I will spend the first 2 weeks of the project familiarizing myself with SCONE, and learning how to run k-eigenvalue MC simulations. Most of my time will be spent preparing for my exams.
- *Exams.*
- The first simulations without FM will be performed in the next 2 weeks.
- *Final exam.*
- Modifications need to be made to the extisting SCONE code for the FM accelerated simulations.
- *University of Pavia trip.*
- *Final coursework deadlines,* I anticipate to have these finished well ahead of time.
- The simulations with FM acceleration are run in June. Depending on my progress, I may perform additional theoretical analysis as discussed in subsection 6.3.
- Theoretical work is finished in the first weeks of July. From now on, any additional time can be spent on extending the research with extra simulations.
- *Presentation abstract deadline.*
- I will start focusing on the presentation starting 1.5 weeks before the conference.
- *Dissertation Conference.*
- Depending on my progress, I will either work on my first draft or spend time extending the research. All research must be finished by August 7.
- *Submit first draft.*
- I give myself almost 2 weeks to finish the report because I want to have plenty for final modifications and extensive proofreading.
- *Dissertation document deadline.*

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APPENDIX: Word Count

1. *Introduction:* 305 words
2. *Monte Carlo Neutron Transport:* 708 words
3. *Neutron Clustering:* 746 words
4. *Fission Matrix Acceleration:* 547 words
5. *Shannon Entropy:* 374 words
6. *Methodology:* 1035 words
7. *Planning:* 276 words

Total: **3991** words